Vertically Integrated Industries

Initial takeaways on minerals, metals, manufacturing and chemistry

1. Investment theses: tailwinds, key takeaways and macro

This theme covers companies that make physical things in their own plants, such as metal parts, magnets and molecules, and that use their own software and AI to make them faster and with fewer skilled people than older competitors. These companies earn money on what they ship. We call them vertically integrated because one company owns the whole job, from taking the order to shipping the product. The five theses below explain why demand is rising, where new companies have an edge, how buyers behave, what the unit economics look like, and where the main risk sits.

1. **The US is rebuilding its supply chains for physical goods, and the main tailwinds behind that will last**
   - China leads refining in 19 of 20 strategic minerals, with about 70% average share (IEA), and makes about 94% of the world's rare-earth magnets, which go into drones, missiles, motors and robots (Mining.com). The US relies fully on imports for 16 critical minerals (USGS), and China has already used export controls on these materials; its current pause on those controls ends on November 10 and November 27, 2026. Buyers and the government will pay for non-Chinese supply even when it costs more, because supply chain resilience is now an economic and national-security issue.
   - Defense demand is rising. The FY27 budget request asks for $49.5B for 10 key munitions, about 8.5 times the average of the past decade, and the Navy is putting $900M into plants that make submarine parts (CNAS, Breaking Defense). The FY26 defense bill lets the Pentagon commit to purchases of up to 10 years with advance payments (FY26 NDAA).
   - The government now acts as a lender and investor. The Pentagon's Office of Strategic Capital (OSC), which lends to companies in key supply chains, has made a $620M loan to one magnet maker and a $500M conditional loan to a rare-earth metals maker, and about $20B is requested for it in FY27 (Mintz). The Department of Energy's loan office has over $289B of lending authority (DOE), and governments worldwide put about $65B into minerals in 2025 (IEA).
   - The labor gap will last. US manufacturers need to fill about 3.8M jobs by 2033, and about 1.9M of those jobs are projected to go unfilled (Deloitte). Of the 17,156 machine shops in the US, 83% have fewer than 20 employees (Census), and many owners are near retirement with no successor, which puts a steady supply of certified shops up for sale.
   - Some tailwinds are temporary, and we should not build the thesis on them. Tariffs are set by proclamation and can change overnight, about 82% of the munitions request depends on a reconciliation bill Congress has not passed (CNAS), and government loans take a long time to close and come with conditions. The Fed raised rates to 3.75-4.00% in September, and spending on new factory construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR). We should favor companies that still work if tariffs fall and the munitions surge slows.
2. **AI and automation give new companies an edge in the work around the machine and in plants that run nonstop**
   - The work around the machine includes quoting, planning, writing the instructions a machine follows (CAM programming), inspection and paperwork. One startup claims it cut the time to write assembly instructions from about 10 hours to 15 minutes (claim), and a former employee of one parts maker said its time from order to first part fell about 50% over two years.
   - The machines themselves sit idle much of the time. In a survey of more than 100 plants, cutting machines ran about 34% of the time against a 70% target, and European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting). Planning and control software can close that gap without a new plant.

- In a plant that runs nonstop, such as a chemical or metals line, every extra point of yield (good output per unit of input) is close to pure profit. Labor is under 10% of the cost of goods in a new automated plant (claim), so the case for automation rests on speed and yield more than on headcount.
- When skilled people are scarce, a plant that needs fewer of them can grow where its competitors cannot. Nobody has shown yet that AI makes each unit cheaper as volume grows, so the edge today is speed, yield and the ability to run with a smaller crew.

3. **Buyers pay for speed and spare capacity, and an approval keeps paying for decades**
   - "Cost is secondary. Cost is driven usually by schedule," a director at a defense prime told us in an expert call, and an engineer there said he would pay 20-40% more to avoid a delay. The premium is highest on prototypes, spare parts and urgent orders, and it fades at volume.
   - A major defense contractor, often called a prime, approves a supplier only after the plant exists and is certified, which takes about nine months to two years, based on what two primes told us in expert calls. Once approved, a supplier keeps the work for aerospace programs that can run about 50 years.
   - The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Most of that supplier base is small, old and short of people, so primes need new suppliers that can deliver on time.
4. **Unit economics work when the company owns the output and someone else carries much of the capital**
   - Owning the plant pays. One listed parts maker earns a 49% gross margin in its own plants and 31% on work it sends to partner shops (company filings). Approved incumbents also earn well: one engine-parts unit runs a 37.7% EBITDA margin (earnings before interest, taxes, depreciation and amortization).
   - Capital spending before contracts are signed is the central risk. Capex means spending on buildings, machines and certifications. One precision-parts maker committed about $200M to a single Arizona plant (Plant Services), its CEO says capex runs about two years ahead of revenue (claim), and process plants need to run at about 80-85% of capacity to make money (FTI Consulting). Equity spent on a building earns building-like returns and dilutes the founders before the first volume order arrives.
   - The companies that succeed shift who pays for the build. They use government loans and grants, buyers that agree to purchase output before the plant is built, customers that host the factory inside their own plants, bank debt once revenue exists, leased buildings, franchising, and acquisitions of plants that are already certified.
5. **Incumbents are the main risk, but the supplier base leaves room for new companies**
   - The primes decide which suppliers are approved and how much work each one gets, and they cap each supplier at about a quarter to a third of their work, so even a winning supplier grows one share at a time. The largest aerospace parts supplier saw its share of one buyer's machining fall from about 50% to 30%, and it is now automating (expert call).
   - Incumbents often buy instead of build. Miners back startups through their venture arms, one miner paid $6.7B for a lithium producer, and a defense contractor's venture arm led a round in an AI materials startup. Drug companies are building their own AI labs, and one has plans to spend up to $1B.
   - Selling hardware to factories has been hard. One 3D-printer maker fell from a $2.1B valuation to a $42.5M sale (3D Printing Journal), and an AI-run machining startup in Germany went insolvent in August 2026.
   - New companies win where the incumbents are weakest: small, aging shops with no successor, new defense programs that need second sources, and plants that need software to run with fewer people.

2. Interesting categories

This section follows the categories in our primer. For each category we set out the main tailwinds and headwinds and then explain why we should or should not focus there, looking at the category as a whole.

A. Raw Materials

Stage A: Locating & finding minerals and mines

- **Tailwinds:** Cutting reliance on China starts with finding new deposits of copper, lithium, nickel and rare earths in the US and allied countries, since the US relies fully on imports for 16 critical minerals (USGS). New mines are needed for batteries, magnets and the grid, and AI combined with cheaper drilling and faster lab testing lowers the cost of checking each target. The major miners are paying for these new methods through their venture arms.
- **Headwinds:** Most finds never earn anything. By one estimate about 0.5% of exploration projects succeed (claim), and only a fraction of the deposits found each year ever become mines. The wait is long: mines average 15.7 years from discovery to production, US mines opened in 2020-24 took 19.1 years, and US permitting alone commonly takes 7-10 years, against about 2 in Australia and Canada.
- **Why we should not focus here:** An explorer that does not own the mine does not own the economics. Explorers usually bring in a miner through an option or joint venture, in which the miner pays staged cash and shares and funds the drilling to earn 51-70% of the project. The explorer keeps a minority stake or a royalty, usually a net smelter return (NSR) of 1-3% of metal sales after transport, and the miner can often buy part of that royalty back at a fixed price (industry option terms). Early ground can sell for a few hundred thousand dollars plus a royalty, and a royalty pays real money only after someone else has built the mine and it is producing (company releases). An explorer that keeps the mine turns into a mining company that needs billions of dollars to build, since a new underground copper mine can cost over $2B (Ecofin), and it then waits 15 to 19 years for production. In both cases the return is too small, too late or too capital-heavy for a venture fund. Selling exploration software to miners has not solved this, because founders who tried it report pilots that did not turn into recurring revenue (claim).

Example companies: Earth AI, KoBold Metals, Terra AI, GeologicAI, Ideon

Stage B: Mining & refining

- **Tailwinds:** This is the stage where China's hold is tightest and where cutting reliance on China matters most. China leads refining in 19 of 20 strategic minerals with about 70% average share (IEA) and makes about 94% of rare-earth magnets (Mining.com), and it has already used export controls on these materials. Drones, missiles, motors, EVs and data centers all depend on that supply, so defense and auto buyers will pay for a non-Chinese source and sign long purchase contracts to secure it. Government support adds to the pull: Pentagon loans and 10-year purchase commitments with price floors, the Department of Energy's $289B of lending authority, and about $65B of government money put into minerals worldwide in 2025 (IEA). Refineries and magnet plants also run nonstop, so AI and automation that raise yield go straight to profit.
- **Headwinds:** Chinese refineries take about 6 months to commission, against 2 to 4 years in the West. A Western plant takes about 5 years to build and another 3 to 5 to ramp after permits (claim), and it costs hundreds of millions of dollars before it earns revenue. Prices for many of these materials are set in China, so a US producer competes against Chinese costs, and the largest US rare-earth producer lost $85.9M on $224.4M of revenue in 2025.
- **Why we should focus here:** Mining and refining sit on the most lasting tailwind in the theme. The need for non-Chinese metal and magnets comes from national security and supply chain resilience, so it does not depend on a tariff or a single budget cycle. Buyers in defense, autos and energy are signing long contracts for this output, which gives a plant years of visible revenue once it runs. New companies have a real edge here: restarting old mines that already have permits shortens the timeline, processing scrap and mine waste avoids a new mine altogether, and AI-run process control raises yield in plants that run around the clock. The category is capital-heavy, so the companies that build here pair venture equity with debt and long purchase contracts, and that financing is part of how the category works.

Example companies: Mariana Minerals, BHP, Rio Tinto, Caterpillar, Vulcan Elements, Phoenix Tailings, ReElement, Ucore, Energy Fuels, MP Materials, Hertha, Cyclic, Mass Magnetics, Duranium, Boston Metal, Electra, Ceibo, Nucor, Carpenter Technology

B. Manufacturing

Metal Supply

- **Tailwinds:** Tariffs and buy-American rules help US metal, and reshoring raises demand from US shops and primes. Job shops want faster quotes and delivery than brokers give them, and online ordering makes that possible.
- **Headwinds:** The business is crowded with established incumbents, and the largest US distributors are gaining share without AI. Distributors earn a spread on the metal plus cutting fees, and much of their cash sits in inventory. Online ordering with instant quotes already exists, so technology does little to set a new distributor apart.
- **Why we should not focus here:** The margin in metal supply stays with the incumbents, who already have the stock, the buying power and the customers. A new distributor ties up its capital in inventory and then competes on price and availability, where the incumbents are bigger. The main tailwind is the tariff, which can reverse, and Part 1 argues against building a thesis on a margin that exists only because of a tariff.

Example companies: Nox Metals, Reliance, Ryerson

Fabrication

- **Tailwinds:** Fabrication means cutting, bending and welding metal into frames and enclosures, and assembling parts like wire harnesses, which are the bundles of wires that carry power and signals through an aircraft or missile. Reshoring and the growth of US hardware companies in defense, drones, space and energy create demand for parts made quickly in the US, and buyers pay for that speed: an engineer at one prime said he would pay 20-40% more to avoid a delay (expert call). The shortage of welders and machinists favors plants that automate cutting and bending and use software for quoting, nesting (packing parts onto a sheet to cut scrap) and scheduling.
- **Headwinds:** The speed premium fades at volume, and US shops beat Chinese prices on only part of their work (claim). Customers that grow often buy their own machines and take the work in-house, and buyers cap each supplier at about a quarter to a third of their work.
- **Why we should focus here:** Fabrication plants cost far less than precision plants, so companies in this category can grow on their own revenue and reach profit early. Owned plants keep more of each sale than shops that route work to partners, and software for quoting and scheduling lets a plant handle many small orders with a small crew. Buyers that pay for speed, low capital needs and early profits make this one of the best fits for a venture fund in the theme.

Example companies: SendCutSend, 1872, Senra Systems, Machina Labs, CircuitHub, Jeevy, Protolabs

Precision & Defense Hardware

- **Tailwinds:** Precision parts are metal or composite parts made to very tight limits, by machining, 3D printing, casting or molding, for missiles, jets, satellites and submarines. National security requires a domestic supplier base the Pentagon can trust, and today it has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Demand is rising, with a $49.5B munitions request for FY27, $900M going into submarine-parts plants and purchase commitments of up to 10 years now allowed. Primes earn 7-12% on cost-plus contracts and outsource much of this work to small shops, and with 83% of US machine shops under 20 employees and many owners retiring, certified capacity is coming up for sale.
- **Headwinds:** A new plant has to be built and certified before it wins volume work. AS9100, the aerospace quality standard, typically takes 6-12 months, and approval at each buyer takes about nine months to two years, so capex runs well ahead of revenue. Much of the munitions money still depends on Congress, and the largest incumbent suppliers are automating.

- **Why we should focus here:** An approved supplier keeps the work for programs that can run 50 years, primes pay a premium for fast delivery, and approved incumbents earn strong margins, so the prize at the end of approval is real. New companies can reach it through four factory models. They can build their own factories, which keeps the whole margin and all the data but means paying for the plant before approval. They can buy certified shops from retiring owners, which comes with machines and approvals already in place, at about seven times annual earnings (claim). They can run a franchised network, where franchisees buy the machines and the company supplies software, orders and quality rules. They can also host cells inside a buyer's plant, which shares the cost with the buyer. Each model trades some margin for less capital, and together they let new companies take share from small, slow shops that cannot keep up with defense demand.

Example companies: Hadrian, Isembard, Divergent, Freeform, Amca, Atomic Industries, Fabri, Layup, Hybron, Rangeview, Furientis, Helicon, Westmag, Re:Build, Anduril, Precision Castparts, Howmet

C. Specialty Chemicals

Producers (hybrid)

- **Tailwinds:** We limit chemicals to automated labs that make new molecules for drug programs. Drug discovery depends heavily on chemistry done in China, and that has become a supply chain, intellectual property and national-security concern. BIOSECURE, a law passed in December 2025, limits the use of certain Chinese biotech suppliers, and 79% of 124 biotechs surveyed had a China-based contract manufacturer (BIO). AI that designs molecules and robots that make them bring down the cost of doing this work in the US and Europe, and UK and Scottish governments are funding lab expansions with grants. Early labs in this category report work with several of the top 20 drug companies (claim, SCI).
- **Headwinds:** US chemists cost several times Chinese rates, so these labs have to automate to reach offshore prices. The category is young, labs have not disclosed revenue, and drug companies are building their own AI labs, with one planning to spend up to $1B.
- **Why we should focus here:** Drug companies want a domestic or allied source for discovery chemistry, and they now have a legal reason to move work out of China. These labs make small batches of new molecules for drug teams at milligram-to-gram scale and do not develop their own drugs. They are paid per compound delivered, or through milestones and royalties if a molecule advances. Each lab costs tens of millions to build, but it owns the software and the reaction data, and that data improves its designs with every run. If automation brings unit cost close to offshore prices, the category can take a meaningful share of work that is leaving China.

Example companies: Chemify, Warp Sciences

D. Advanced Materials

Producers

- **Tailwinds:** China makes about 85% of battery cathode material and over 98% of the cheaper LFP type, so battery makers for EVs, grid storage and defense need another source for resilience and security. Data centers used 415 TWh of power in 2024 and are on track for about 945 TWh by 2030, with cooling taking 7% to over 30% of that power, which creates demand for new cooling materials. One AI-designed cooling fluid reached market in about 3 years, against roughly 10 years and $100M for a typical new material (claim), and contract manufacturers let a company make a new material without building its own plant.
- **Headwinds:** Customers take one to five years to qualify a new material. Owned cathode plants cost hundreds of millions of dollars and compete on Chinese prices, and an established incumbent's battery-materials unit lost money on €436M of revenue.
- **Why we should focus here:** These companies design the material with AI in their own lab, then make it and sell it by the kilogram or ton under supply deals, so the company that designs and makes the material keeps the margin. The demand for non-Chinese battery materials and for better cooling comes from lasting

needs in energy, defense and data centers. Faster AI design shortens the development time, and contract manufacturing keeps the capital need far below that of a mine or a refinery.

Example companies: Orbital Materials, Mitra Chem, Chemix, Albemarle, Umicore, Corning

Materials Discovery Labs

- **Tailwinds:** AI and robotic labs can test far more candidates than people can, and some have made more than a thousand alloys in a few months (claim). The same needs that drive the rest of the theme, from non-Chinese magnets and battery materials to better alloys for defense and cooling for data centers, create demand for new materials. Government research awards and partner R&D budgets already fund this work.
- **Headwinds:** These labs do not yet sell material at volume, and finding a material faster does not shorten the years customers take to qualify it. Revenue today comes from paid R&D, licensing, software and government research awards of around $1M, and valuations in the category are high relative to that revenue. One investor says materials has had "no AlphaFold moment" yet.
- **Why we should focus here, on a longer time horizon:** If these labs work, they shorten the slowest part of the materials chain, which is finding a material worth qualifying. To sell at volume they usually go through a partner that scales the material up and sells it, so a lab earns a fee or royalty unless it later builds its own production. The category is more of an experiment today, and we should expect returns on a longer timeline than in the producer categories.

Example companies: Radical AI, CuspAI, Lila Sciences, Periodic Labs, Aionics, Mattiq, Dunia Innovations, Entalpic, Discovered Materials, Kebotix, Citrine Informatics, Uncountable, Albert Invent, Microsoft Discovery

E. Enabling Software & Robotics

Enablers

- **Tailwinds:** Reshoring only works if US plants can run with fewer skilled people, and about 1.9M US manufacturing jobs are projected to go unfilled by 2033 (Deloitte). The US also needs to catch up on automation, having installed 38,428 industrial robots in 2025 against 354,226 in China. Existing plants leave a lot of capacity unused, since cutting machines run about a third of the time against a 70% target and European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting). AI is making the work around the machine much cheaper, cutting tasks like writing assembly instructions from hours to minutes (claim).
- **Headwinds:** Sales to plants are slow, and incumbent automation vendors bundle software with the hardware they have already installed. Robots are not yet reliable enough for most factory work, since robot-learning success is 59-66% on short tasks against the 99.9% or more that a factory needs.
- **Why we should focus here:** There is no factory to build, so the money goes into product and sales, which fits a venture fund. Plant-control software adjusts settings such as a temperature or flow rate on its own and can be paid on the extra production it delivers, factory software is sold by subscription, and robot makers sell or rent robots and robot cells. In a sold-out plant each extra point of yield is close to pure profit, so buyers can measure the payback. Most other categories in this memo need these tools to run with fewer people.

Example companies: Lumaril, Arda, Mind Robotics, GrayMatter Robotics, CloudNC, AI Robotics, Bright Machines, Rapta, Allus, Faraday, Siemens, Rockwell Automation, Honeywell, AspenTech, Emerson